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Network Tomography Based on Additive Metrics

Authors Jian Ni, Sekhar Tatikonda

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Domain Networking & distributed systems

Abstract

Inference of the network structure (e.g., routing topology) and dynamics (e.g., link performance) is an essential component in many network design and management tasks. In this paper we propose a new, general framework for analyzing and designing routing topology and link performance inference algorithms using ideas and tools from phylogenetic inference in evolutionary biology. The framework is applicable to a variety of measurement techniques. Based on the framework we introduce and develop several polynomial-time distance-based inference algorithms with provable performance. We provide sufficient conditions for the correctness of the algorithms. We show that the algorithms are consistent (return correct topology and link performance with an increasing sample size) and robust (can tolerate a certain level of measurement errors). In addition, we establish certain optimality properties of the algorithms (i.e., they achieve the optimal l-infinity-radius) and demonstrate their effectiveness via model simulation.

Real-World Impact & Application

Borrowing Biology Algorithms to Map the Internet: Ni & Tatikonda discovered a wild connection: packet lag traveling across internet cables follows the exact same mathematical rules as genetic mutations accumulating over generations. Because of this paper, internet engineers can use 40 years of evolutionary biology software to map computer networks!